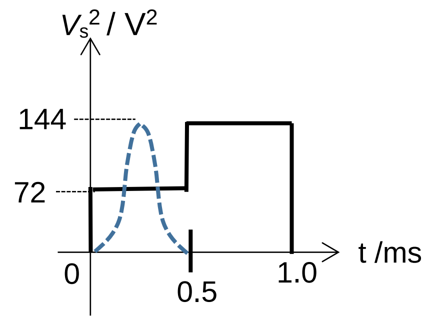
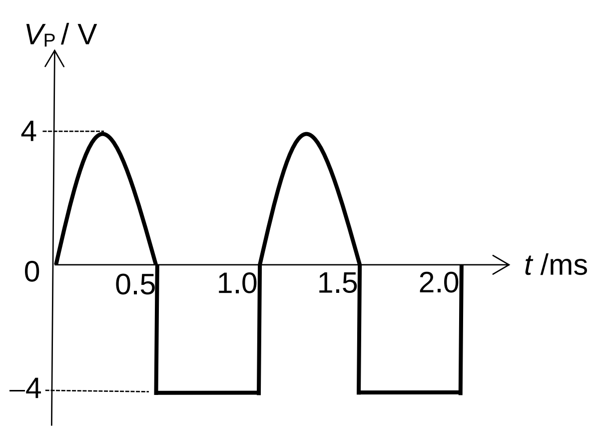
<table>
<tr><td>5</td><td>(a)</td><td>The value of voltage that is <u>equivalent to that of a steady d.c. voltage</u> which will dissipate energy <u>in a given resistor at the same rate</u>.</td><td>B1
B1</td></tr>
<tr><td></td><td>(b)(i)</td><td>Squaring the graphs and arranging the curved portion to a rectangle.


Mean-square-voltage = 108 V^2
Root-mean-square voltage = 10.39 V = 10.4 V</td><td>M1

M1
A1</td></tr>
<tr><td></td><td>(b)(ii)</td><td><table><tr><td>Diode 1</td><td>Forward-biased</td></tr><tr><td>Diode 2</td><td>Reverse-biased</td></tr></table>
Both correct – [1]
One or zero correct – [0]</td><td>B1</td></tr>
<tr><td></td><td>(b)(iii)</td><td>Effective resistance across resistors P and Q = 10/2 = 5 Ω
Using potential divider rule,
Max p.d. across P = $\frac{5}{10+5}(12) = 4.0$ V</td><td>B1

B1</td></tr>
<tr><td></td><td>(b)(iv)</td><td>

Correct waveform (shape) for two cycles
–12 V < V_p < 12 V
Numerical values of V_P not assessed</td><td>B1
B1</td></tr>
</table>

6	(a)(i)	either wavelength is longer than threshold wavelength	